

12.164

EE25BTECH11047 - RAVULA SHASHANK REDDY

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**Question:**

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find the maximum number of linearly independent eigenvectors of  $\mathbf{A}$ .

**Solution:**

Since  $\mathbf{A}$  is a triangular matrix

$$\lambda = 1, 1, 1 \tag{1}$$

$$(\mathbf{A} - \mathbf{I})\mathbf{x} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{x} = 0, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \tag{2}$$

$$y + z = 0 \implies y = -z \tag{3}$$

$$z = 0 \implies y = 0 \tag{4}$$

$$\mathbf{x} = \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \tag{5}$$

Maximum number of linearly independent eigenvectors = 1